

UEI301: DIGITAL ELECTRONICS				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: To familiarize the student with the analysis and design of various digital systems.				
Syllabus				
Introduction: Difference between analog and digital systems, Advantages and Disadvantages of digital system. Binary codes: Weighted and non-weighted codes, Sequential codes, Self-complementing codes, Excess-3 code, Gray code, Error-detecting codes, Error-correcting codes, Hamming code.				
Realization of Logic Gates Using Diodes & Transistors: AND, OR and NOT Gates using Diodes and Transistors, DCTL, RTL, DTL, TTL, CML and CMOS Logic Families and its Comparison, Classification of Integrated circuits, comparison of various logic families, standard TTL NAND Gate Analysis & characteristics, TTL open collector O/Ps, Tristate TTL, MOS & CMOS open drain and tristate outputs, CMOS transmission gate, IC interfacing- TTL driving CMOS & CMOS driving TTL.				
Sequential Circuits: Basic Bistable element, Latches, SR latch, Application of SR latch, A Switch debouncer. The SR latch, The gated SR latch. The gated D Latch, The Master-Slave Flip-Flops (Pulse- Triggered Flip-Flops): The master-slave SR Flip-Flops, The master- slave JK Flip-Flop, Edge Triggered Flip-flop: The Positive Edge-Triggered D Flip-Flop, Negative-Edge Triggered D Flip-Flop. Characteristic equations, Registers, Counters- Binary Ripple Counter, Synchronous Binary counters, Counters based on Shift Registers, Design of a Synchronous counters, Design of a Synchronous Mod-N counters using clocked JK Flip Flops Design of a Synchronous Mod-N counter using clocked D, T, or SR Flip-Flops				
Sequential Systems: Introduction, Mealy and Moore models, State machine notation, synchronous sequential circuit analysis and design. Construction of state Diagrams, Counters Design.				
Asynchronous Sequential Logic: Analysis Procedure, Design procedure, reduction of state and flow table, race free state assignments, hazards, Design of Asynchronous sequential circuits.				
Converters: Digital to Analog conversion, R-2R ladder DAC, Weighted Resistor DAC, Analog to Digital (A/D or ADC) conversion, Flash type, Counter type ADC, Dual-slope ADC, Successive approximation type ADC, sigma-delta ADC.				
Memories: Memory Units, Memory Addressing, Introduction and classification of ROM, Static and Dynamic RAM, Flash memory, Memory Expansion, FIFO Memory, LIFO				

Memory.

Introduction to HDL: Introduction, A brief history of HDL, Structure of HDL Module, Operators, Data types, Types of Descriptions, Simulation and synthesis, Brief comparison of VHDL and Verilog. Data Flow Descriptions: Highlights of Data flow descriptions, Structure of data-flow description, Data type- vectors.

Laboratory Work: To consider various important codes and the logic for converting from one to another, Understanding of ICs 74146, 7476, 7483, 7485, 7490, 7492, 7495, 74121, 74123, 74126, 74151, 74163, 74180, 74181, 74190, 74192, 74195, 74196, Shift register and binary counting using JK flip flop, asynchronous/synchronous up/down counters, Variable modulus counters, HDL and VHDL programming.

Course Learning Objectives (CLO)

The students will be able to:

1. Differentiate between different number systems and various codes
2. Design the asynchronous sequential Logic circuits
3. Exhibit the knowledge of different analog to digital converters, logic families and realization of logic gates
4. Elucidate the concept of counters and sequence generators memories and logic circuits

Text Books

1. Digital Fundamentals, Floyd, T.L. and Jain, R. P., Pearson Education (2008).
2. Digital Systems: Principles and Applications, Tocci, R. and Widmer, N., Pearson Education (2007).
3. HDL Programming (VHDL and Verilog), Nazeih M. Botros Cengage Learning 1 st Edition, 2011.

Reference Books

1. Digital Design, Mano, M. M. and Ciletti, M., Pearson Education (2008).
2. Fundamentals of Digital Circuits, Kumar, A., Prentice Hall (2007).
3. Digital Logic Design and VHDL, A.A.Phadke S.M.Deokar Wiley India 1st Edition, 2009
4. Digital Circuits and Design, D.P.Kothari, J.S.Dhillon, Pearson First Print 2015
5. Circuit Design and Simulation with VHDL, Volnei A Pedroni PHI 2nd Edition

Evaluation Scheme	
Evaluation Elements	Weightage %
Mid Semester Test (MST)	25-30
End Semester Examination (ESE)	40-45
Sessional (may include Assignment, Sessional (Includes Regular Lab assessment) and Quizzes Project (Including report, presentation etc.)	30